MANUFACTURING COST MANAGEMENT

Would you fund this factory improvement?

A liquid detergent line investment case

Informed by Unilever’s public reporting.
Built around a hypothetical UK production line.

Sankha Bhattacharjee

Independent cost management portfolio project

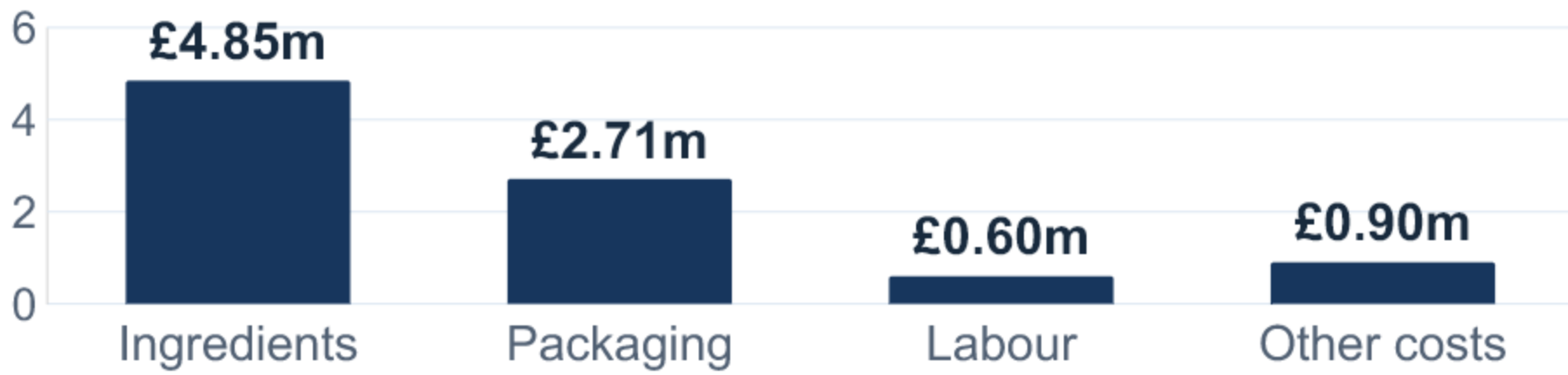
Illustrative study. Costs and savings are unvalidated.

Costs begin with quantities and rates

Assumed annual output: 10 million one-litre bottles

Liquid mass balance	Annual kg
Material input	10,618,557
Acceptable product	10,300,000
Process loss at 3%	318,557

Baseline annual production cost (GBP millions)



£0.907 baseline production cost per bottle

The investment and its assumptions

Initial economic cost	GBP
Equipment, installation, validation and contingency	£92,000
Pilot and training	£10,000
Lost contribution at installation	£15,767
Total	£117,767

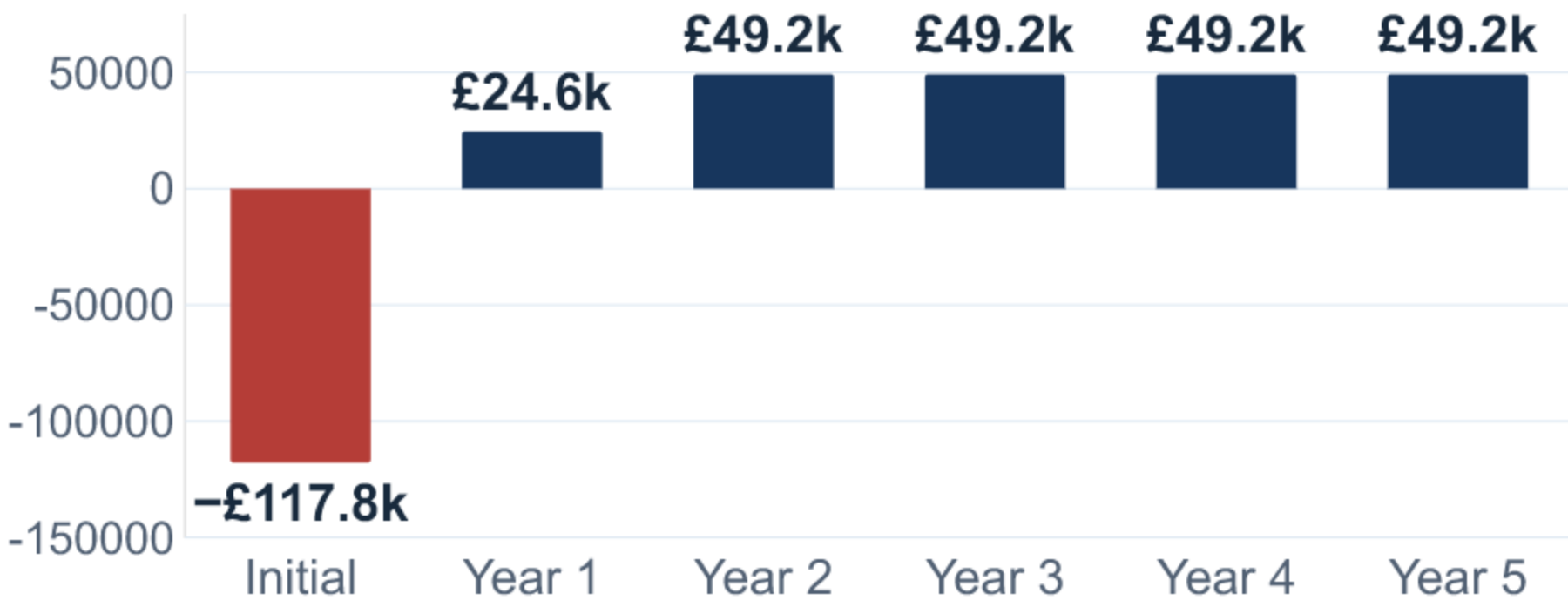
3% to 2%

Assumed process loss as a share of input

Maintenance: £8,000 per year
Savings begin in July. Payroll stays fixed.

The base case creates long-term value

Incremental economic cash flow (GBP)



£49,205

Annual net cash saving

£46,392

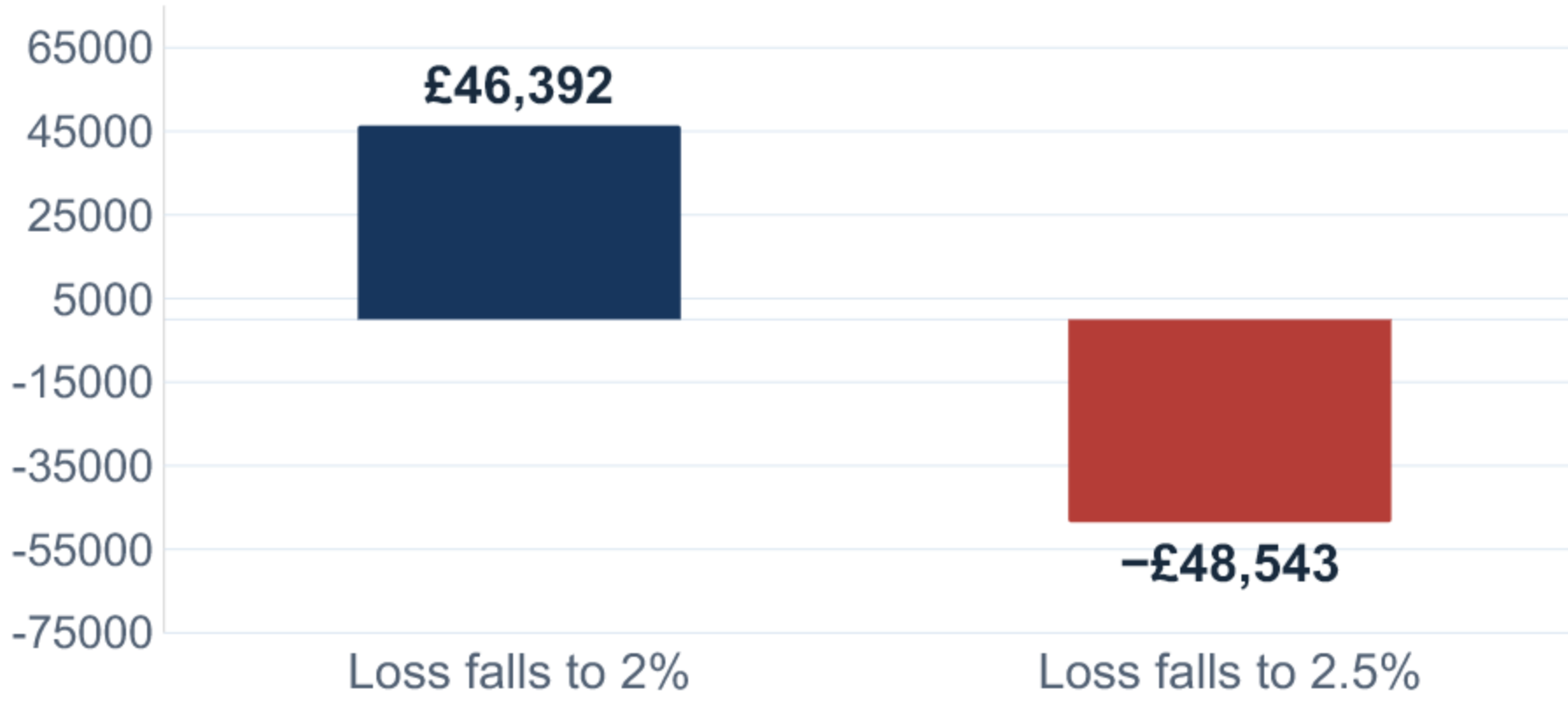
Five-year NPV at 10%

Year 1 operating profit falls by £10,364.
Upfront expenses and delayed savings affect the first year.

Illustrative study. Costs and savings are unvalidated.

A smaller gain reverses the decision

Five-year net present value (GBP)



Break-even process loss

Approximately 2.245% or lower

At the current budget and other model assumptions.

Illustrative study. Costs and savings are unvalidated.

My recommendation: a measured pilot

01 Establish the baseline

Measure input mass, acceptable output and actual losses.

02 Test the improvement

Compare matched batches and verify product quality.

03 Confirm the complete cost

Obtain equipment quotes and validate energy and upkeep.

04 Review the investment

Use measured net savings to decide whether to proceed.

What evidence would you require before approval?

Public context: Unilever’s 2026 digital twins release

Illustrative study. Costs and savings are unvalidated.